**Full Stack Assignment – 34**

**Assignment ID:** FSD-34

**Title:** Access Revocation Management System

**Duration:** 3 Hours

## 1️⃣ Objective

Design and develop an **Access Revocation Management System** that allows administrators to grant and revoke user access while ensuring revoked access cannot be self-restored.
The system must enforce revocation rules and ensure **data persistence using a database**.

## 2️⃣ User Roles

**USER**

- Register and log in to the system
- View current access status

**ADMIN**

- Register and log in to the system
- Grant access to users
- Revoke access from users
- View access history

## 3️⃣ Functional Requirements

**Authentication & Authorization**

- Implement user registration and login
- Use **JWT-based authentication**
- Protect all backend APIs using JWT
- Enforce **role-based access control** on APIs and frontend views

**Access Management**

- ADMIN can grant or revoke user access
- Access status must be stored in the database
- USER access status must be retrieved dynamically

## 4️⃣ Business Rule (Critical)

⚠️ **Revoked access must not be restorable by the user.**

Once access is revoked, only an ADMIN can restore it.

## 5 Technical Constraints (Mandatory)

- All access and revocation data **must persist in a database**
- Hardcoded or in-memory storage is **not allowed**
- Backend APIs must reflect actual database state
- JWT authentication and role validation are mandatory
- Use environment variables for sensitive configuration

## 6 Frontend Requirements

- Login and registration pages
- Role-based dashboards for USER and ADMIN
- Access status view for USER
- Access grant/revoke management view for ADMIN

UI design is not evaluated; authorization correctness is the priority.

## 7 Submission Requirements (Strict)

**GitHub Repository**

- Backend source code
- Frontend source code
- Proper folder structure
- Clear and meaningful commit history

**Live Deployment**

- Live frontend deployment link
- Live backend deployment link
- Backend must be connected to the database

**README.md (Mandatory)**

The README file must include:

- Project overview
- Tech stack used
- User roles and permissions
- API endpoints
- Database schema or collections
- Steps to run the project locally
- Live frontend and backend links

## 8 Evaluation Note

Evaluation focuses on **revocation enforcement, role control, and end-to-end execution**.